

# pyALDIC Quick Guide

Two-page reference. Full walkthrough: `user-guide.tex`.

## 1. Install & launch

Pick one install path (all are equivalent at runtime):

- **From PyPI** (easiest): `pip install al-dic`.
- **From a GitHub Release wheel**: download `al_dic-*.whl` from [github.com/zachong/pyALDIC/releases](https://github.com/zachong/pyALDIC/releases) and run `pip install al_dic-*.whl`. Useful when behind a firewall that blocks PyPI.
- **From source** (development): `git clone https://github.com/zachong/pyALDIC.git` then `pip install -e ".[dev]"` inside the clone.

Requires Python  $\geq 3.10$ . Launch: `al-dic` (or `python -m al_dic`).

## 2. Load images

Drop an image folder onto the drop zone or click Browse. All files must be the same size. Frame 1 is always the reference. Frames stream from disk on demand, so hundreds of 4K images load without exhausting RAM.

## 3. Choose Workflow Type

This decides which frames need a Region of Interest, so do it *before* drawing.

| Experiment                    | Tracking + Ref Update |   |        |
|-------------------------------|-----------------------|---|--------|
| Soft material <2% strain      | Accumulative          |   |        |
| 2–30% strain, any rotation    | Incremental           | + | Every  |
| >5°, impact                   | Frame                 |   |        |
| Known quasi-static load steps | Incremental           | + | Custom |
|                               | Frames                |   |        |

Solver: **AL-DIC** default (smoother,  $\sim 3\times$  slower); **Local DIC** for quick iteration.

## 4. Draw Region of Interest

The blue-accent hint above the toolbar tells you which frames need a region. Toolbar:

- **+** **Add**, **Cut**, **+** **Refine** (brush, frame 1 only). Each opens a Polygon / Rectangle / Circle menu.
- **Import** / **Batch Import** – load PNG masks.
- **Save** / **Invert** / **Clear** – per-frame mask operations.

For per-frame editing, click the **Need** / **Edit** / **Add** button on a row in the image list.

## 5. Parameters

| Control      | Guidance                                                                                        |
|--------------|-------------------------------------------------------------------------------------------------|
| Subset Size  | 41 default; larger (51–81) for noise, smaller (21–31) for fine speckle.                         |
| Subset Step  | 8 for $512^2$ , 16 for $1024^2$ . Halving step $\rightarrow \sim 4\times$ runtime.              |
| Search Range | 20 default; raise for large inter-frame motion. Auto-expand is on.                              |
| Refinement   | Check Inner / Outer to densify mesh near hole or ROI boundaries; pick a level (Light to Ultra). |

**Advanced** section (collapsed by default) holds the initial-

guess mode and auto-expand toggle; defaults are fine.

**Starting Points** (special init-guess mode). Select this radio for large displacement ( $> 50$  px) or discontinuous fields (cracks, shear bands). Every connected mask region shows as **yellow** on the canvas; place at least one point per region (*Place Starting Points* to click, or *Auto-place* to pick the highest- NCC node per region) until every region turns **green**. Run stays disabled until that's true. On a reference-frame switch the points are warped automatically; if warping fails the pipeline auto-places fresh points on the new reference and continues.

## 6. Run & Cancel

Click the blue **Run DIC Analysis**. Progress, elapsed / remaining, and console log appear on the right. Red **Cancel** stops cleanly (partial results kept).

Fatal errors raise a modal dialog; the full traceback is in the console log.

## 7. View results

On the right sidebar after a run:

- **FIELD**: Disp U / Disp V toggle. *Show on deformed frame* checkbox sits here (controls WHERE, not how).
- **VISUALIZATION**: Colormap, Auto / Fixed range radio pair (Fixed = manual Min / Max), opacity. Numeric boxes accept both “.” and “,” decimals.
- **PHYSICAL UNITS**: Pixel size + frame rate – scales displacement / velocity output; strain is dimensionless.

## 8. Strain post-processing

A separate **Strain window** opens automatically when the run finishes.

- **Method**: *Plane fitting* (smooth, default) vs *FEM nodal* (higher resolution, noisier).
- **VSG size**: Virtual strain gauge diameter (default 41 px).
- **Smoothing**: Off / Light ( $0.5\times$  step) / Medium ( $1\times$  step) / Strong ( $2\times$  step) ▲.
- **Strain type**: **Infinitesimal** (small deformation only), *Eulerian-Almansi*, **Green-Lagrangian** (use for any rotation  $> 2^\circ$ ; it is exactly zero under rigid rotation).

Click green **Compute Strain** whenever parameters change. Field selector now exposes **strain\_exx**, **strain\_eyy**, **strain\_exy**, principal strains, **strain\_maxshear**, **strain\_von\_mises**, **strain\_rotation**, plus the displacement fields.

## 9. Export

Both windows have an **Export Results** button; they open the same 5-tab dialog:

- **Data**: NPZ / MAT / CSV of any selected fields.
- **Images**: per-field JPEG (quality 60–100) / PNG / TIFF per frame; resolution presets cap the long edge (512–2048 px or full). Colorbar ON by default.
- **Animation**: MP4 / GIF, encoded frame-by-frame (no

RAM spike); *Frame step* keeps every  $N$ th frame with playback duration preserved. Requires `ffmpeg` on PATH for MP4 (falls back to GIF otherwise).

- **Report:** multi-page PDF summary of the run.
- **Preview & Colorbar:** WYSIWYG preview through the real export path. Colorbar position / font / thickness / background / margin; field colormap-range-opacity synced two-way with Images; *Apply to all fields*.

## 10. Save / resume sessions

**File** → **Save Session** (`Ctrl+S`) writes a single `.aldic` bundle: every Region of Interest, all parameters, view state, and — optionally, a prompt shows the size estimate — the computed results. Async, with a progress dialog.

**File** → **Open Session** (`Ctrl+O`) restores the exact page (frame, field, colormap, ranges) with no recompute. Legacy `.aldic.json` files still open. If the image folder is gone, parameters and Regions of Interest still load; re-drop the images manually. Starting Points are not saved — re-place them before the next Run.

Windows: **File** → **Associate .aldic files** makes double-clicking a session file open it in pyALDIC.

## 11. Troubleshooting quick table

| Symptom                              | Likely cause / fix                                                                                                                         |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Run button disabled                  | No images or no Region of Interest on frame 1.                                                                                             |
| Crazy strain under big rotation      | Using Infinitesimal strain; switch to Green-Lagrangian.                                                                                    |
| Strain map / export all blank or NaN | Plane-fit edge-trim removed every node — ROI too thin or VSG too large. Lower <i>Trim</i> ( $\alpha$ ), reduce VSG size, or use FEM nodal. |
| FFT peaks clipped warnings           | Raise Search Range or keep auto-expand on.                                                                                                 |
| Refine brush grayed out              | Brush is frame-1 only; navigate to frame 1.                                                                                                |
| Results poor near hole edges         | Enable <i>Refine Inner Boundary</i> .                                                                                                      |
| Slow run                             | Halve Subset Step for next iteration; use Local DIC for pre-views.                                                                         |
| Session won't load                   | Check console: corrupt file, future schema, or missing image folder (warning only – params restore).                                       |